

LAKE WATEREE, SC, USA

WMO: 998208

Lat: 34.335N

Lon: 80.702W

Elev: 87

StdP: 100.28

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -3.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 8.4	(k) 7.1	(l) 7.7	(m) 7.3	(n) 1.3	(o) 140	(p) 0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 36.8	(d) N/A	(e) 35.4	(f) N/A	(g) 34.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.1	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.8	(b) 5.4	(c) 4.5	(e) -9.2	(f) 38.7	(g) 2.6	(h) 1.7	(i) -11.1	(j) 39.9	(k) -12.7	(l) 40.9	(m) -14.1	(n) 41.8	(o) -16.0	(p) 43.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.4	8.7	12.0	17.7	22.3	26.4	27.6	26.7	24.2	17.7	11.2	9.5		
	DBStd	8.38	5.09	5.02	4.74	3.70	3.22	2.41	1.96	2.06	2.88	3.86	4.03	4.88		
	HDD10.0	353	137	79	32	1	0	0	0	0	0	1	32	70		
	HDD18.3	1461	371	272	204	55	8	0	0	0	1	57	216	276		
	CDD10.0	3119	25	42	93	233	381	492	544	519	425	241	68	55		
	CDD18.3	1184	1	2	7	36	131	242	286	261	176	38	2	2		
	CDH23.3	11675	3	17	90	417	1282	2526	2958	2454	1539	371	14	4		
	CDH26.7	5473	0	1	15	101	526	1317	1516	1202	696	99	0	0		
(14) Wind	WSAvg	1.6	1.8	1.8	1.9	1.8	1.5	1.4	1.4	1.2	1.3	1.5	1.6	1.6		
(15) Precipitation	PrecAvg	1164	113	91	110	74	76	112	128	122	94	80	75	82		
	PrecMax	1565	229	190	258	167	184	261	252	274	224	430	217	204		
	PrecMin	778	18	21	10	4	16	16	11	14	6	0	8	10		
	PrecStd	185	51	46	51	44	42	65	70	62	57	78	45	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	25.7	28.7	31.2	35.8	38.7	38.7	37.3	36.0	33.5	25.1	24.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	19.8	22.8	25.8	29.3	33.7	36.4	36.9	35.4	34.2	29.0	23.1	21.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	17.9	20.4	23.7	27.8	31.5	34.7	35.3	34.1	32.7	27.1	21.1	20.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	15.4	18.0	21.2	25.9	29.8	33.2	33.6	32.6	31.0	25.2	19.2	18.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		12.3	12.6	13.2	13.8	12.7	12.6	11.8	11.3	11.9	13.1	13.7	11.9		
	5% DB	MCDBR	15.0	15.1	16.5	16.3	15.4	15.0	14.4	13.6	14.9	15.8	16.4	12.5		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.331	0.368	0.414	0.475	0.504	0.555	0.526	0.443	0.397	0.346	0.318		
	taud		2.495	2.463	2.389	2.248	2.128	2.088	1.968	2.039	2.255	2.352	2.457	2.539		
	Ebn at Noon		889	917	905	874	821	794	751	766	818	829	850	873		
	Edh at Noon		85	97	114	137	156	163	182	167	129	108	88	77		
(44) All-Sky Solar Radiation	RadAvg		2.74	3.37	4.46	5.59	6.13	6.42	6.07	5.52	4.75	3.94	3.01	2.38		
	RadStd		0.14	0.36	0.33	0.42	0.49	0.45	0.37	0.26	0.41	0.48	0.24	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+130

Nomenclature: See separate page